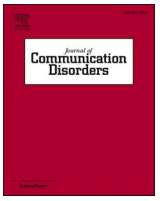
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Calibration matters: II. Measurement of ambient noise in test rooms/areas

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ABSTRACT

Ambient noise measurement is a part of audiometric calibration in which one measures the ambient noise level in a sound room/test area intended for audiometric testing and then decides whether the background noise in the test room meets the maximum permissible ambient noise level (MPANL) requirements specified in national or international standards, e.g., ANSI/ASA S3.1:1999(R2018) or ISO 8253–1:2010 (R2021). If the ambient noise levels are below the MPANLs, clinicians can be sure that the test stimuli they present to patients are not masked by the background noise in the test room/area and their test results are valid and the subsequent clinical decisions are sound. Audiometric testing, however, may not always be carried out in sound rooms/test areas with ambient noise levels below the MPANLs, especially during community outreach or humanitarian services. A thorough understanding on the MPANL requirements for different transducers can help clinicians determine which equipment is appropriate for the test area. This tutorial discusses the rationale and assumptions behind the MPANL specifications, how to measure ambient noise levels of test rooms/areas, and how to apply the national and international standards to determine if the test room is suitable for audiometric testing. Alternative strategies are discussed when the ambient noise levels exceed the specified MPANLs. The rationale and procedures are explained using examples on how to lower the ambient noise levels in test areas, and how to determine the suitable test frequency range and the lowest threshold levels that can be assessed in the test area.

1. Introduction

The measurement of ambient noise levels in a sound room/test area and the determination of the suitability of the test area for audiometric testing are integral parts of the audiometric calibration process because noise in the testing environment can potentially mask the pure tone stimuli and increase the listener's hearing thresholds. As audiometric tests aim to search the lowest sound pressure level that a person can detect across a range of frequencies (e.g., between 250 and 8000 Hz, or to 12,000 and 16,000 Hz in high-frequency audiometry), the ambient noise needs to be kept below the maximum permissible ambient noise levels (MPANLs) specified in national and international standards to prevent masking of the test signals.

The discussion of the importance of MPANLs has been a popular topic in industrial audiology because the hearing thresholds of

Abbreviations: AC, air conduction; ANSI, American National Standards Institute; ASA, Acoustical Society of America; IEC, International Electrotechnical Commissoin; ISO, International Organization for Standardization; MPANL, maximum permissible ambient noise level; RETSPL, reference threshold sound pressure level; SLM, sound level meter.

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people working in noisy environments are required to be documented annually and most of the tests are conducted in the field or onsite. The ambient noise levels measured in many mobile vans or single-walled sound booths may be higher than the MPANLs specified in ANSI/ASA, S3.1:1999(R2018) or ISO 8253-1: (2010) (R2021) (Frank, 1994; Lankford et al., 1999).

Testing thresholds in environments with ambient noise higher than the MPANLs can result in artificially higher thresholds (Frank & Williams, 1993). Nevertheless, not all agencies/organizations require the test room to meet the national or international standards. For example, Occupational Safety and Health Administration (OSHA, 2021) allows higher MPANLs in testing environments which do not allow the testing of hearing thresholds down to 0 dB HL (Berger, 2006; OSHA, 2021). In other cases, such as in humanitarian service settings, it may be impossible to keep the ambient noise below the MPANLs. It is, therefore, important to understand the rationales behind the MPANL standards, the consequences of not meeting the standards, the ramifications on testing accuracy and clinical implications, and methods to lower ambient noise in test rooms.

The MPANLs specified in national and international standards are generally dependent on 1) the average hearing thresholds of young otologically normal individuals, 2) the characteristics of the transducers and their cushions (if any) used to test hearing thresholds, 3) the test frequency range, and 4) whether the ambient noise levels are measured in the octave or 1/3 octave bands. As the International Standards Organization (ISO) and American National Standards Institutes and Acoustical Society of America (ANSI/ASA) acquired their average hearing thresholds from otologically normal adults ages 18–25 years in separate occasions testing different populations (ISO 389-3:2016 Parts 1-8; ANSI S3.6), their normal hearing thresholds and MPANLs in test rooms are slightly different (ANSI/ASA S3.1, 1999 (R2018), ISO 8253-1: (2010) (R2021)).

As different transducers and their cushions (if any) have different abilities to reduce the amount of noise getting into the ear canal (Margolis & Madsen, 2015; Smull et al., 2019), the MPANLs are specific to the transducers used to deliver the test stimuli in "ear covered" conditions (e.g., insert earphones, supra-aural headphones, or circumaural headphones). The MPANLs also specify the ambient noise levels for "ears uncovered" conditions, in which either or both ears are not covered by any transducer(s), such as during bone conduction (BC) and sound field threshold testing.

Additionally, ANSI/ASA and ISO assume the effect of upward spread of masking is 14 dB/octave (ANSI/ASA S3.1:1999 (R2018)). Upward spread of masking is a phenomenon that a louder low frequency sound creates a masking effect to make sounds with higher frequencies less audible (Egan & Hake, 1950). Both standards specify the MPANLs from 125 – 8000 Hz for different transducers and different test frequency ranges (e.g., from 125 to 8000 Hz, from 250 to 8000 Hz). The MPANL requirements are different at low frequencies for different test frequency range when the effects of upward spread of masking are taken into account.

Further, MPANLs are specified in octave and 1/3 octave bands in both ANSI/ASA and ISO standards. The MPANL values for the octave filter measurements are higher than those for 1/3 octave band filters because octave bands have wider bandwidths and measure the levels of a wider range of ambient noise than 1/3 octave bands.

Taken all these factors into account, both ANSI/ASA and ISO standards specify the MPANLs for different transducers with their cushions (if any), different test frequency range, and the filter bandwidth along with their corresponding MPANLs for testing down to 0 dB HL at different frequencies. When determining whether a test area is suitable for audiometric testing, one needs to pay attention to all these factors and apply the appropriate MPANLs.

While the need for a quiet environment to conduct hearing tests is universally understood among audiology professionals, issues surrounding MPANLs and methods to resolve problems related to MPANLs are often less well-understood. A thorough understanding on MPANLs, however, not only can help clinicians determine whether their test results are valid, but also allow them to determine whether the equipment used for audiometric tests are appropriate. This knowledge is especially important for testing hearing in the field where sound rooms are not available.

During community outreach or humanitarian services, for example, tests may be conducted in activity rooms, classrooms, or in open space. Sometimes, the testing site(s) may be a busy clinic beside a busy street. The ability to measure the ambient noise levels and the awareness of what alternative strategies can be employed to increase the validity of the test results are essential.

Insert earphones are commonly used in high-income countries because they have higher noise attenuation ratings, higher inter-aural attenuation, and more comfortable than supra-aural headphones (Killion et al., 1985; Clark & Roeser, 1988; Munro & Agnew, 1999; Gross, 2005; Munro & Contractor, 2010; ANSI/ASA S3.1:1999(R2018); ISO 8253-1: (2010) (R2021); Yacullo, 2015). They can also prevent collapsed ear canals during air conduction testing (Oda et al., 2014). In mid- to low-income countries, however, the cost of insert foam tips can be as high as $1 United States dollar per tip (e.g., in Brazil). Insert earphones, therefore, are rarely used because the cost is prohibitive given the countries' income levels. To circumvent the high cost, some audiology clinics develop methods to clean the foam tips to make them reusable (e.g., cleaning them using heat and disinfectant or surgical spirit). Sometimes, a hearing clinic/testing site is located beside a busy street and the testing area can be contaminated by noises from the street and the surrounding areas. While reusing foam tips are not encouraged, when should a clinic consider investing in cleaning methods if the ambient noise cannot be reduced to acceptable levels? What are other alternatives? When the investment of transducers with even higher noise attenuation ratings is warranted?

MPANLs are usually discussed as part of calibration in textbooks and some articles (Decker & Carrell, 2004; Margolis & Madsen, 2015; Silman et al., 2022; Wilber & Burkard, 2015). In-depth discussions on what SLM settings should be used for ambient noise measurements, how to handle situations when ambient noise level exceeds the MPANLs, and options and management plans are often lacking. This is partly because the readers of the books and articles likely have certain levels of understanding on how to use SLMs, acoustics, and calibration concepts and the authors wrote the books and articles accordingly. Also, in high-income countries, hearing tests are conducted in sound rooms and calibration is carried out by companies that offer annual services. Clinicians usually do not need to worry about the MPANLs and how to resolve the problems related to ambient noises. For humanitarians, however, ambient noise is a constant threat when testing in the field. It is also in forefront of their considerations when they help others set up an

audiology clinic in foreign countries.

As audiology educational programs and calibration equipment are sparsely available in some parts of the world (e.g., Africa and South America), this tutorial and this series of tutorial aim to discuss calibration issues and practices from a basic level. The goal is to introduce the calibration concepts and practices and to build up the calibration capacity in audiology communities, especially in low- to mid-income countries. Examples, figures, and tables are used to illustrate key concepts and procedures. The first tutorial discusses basic acoustics and calibration concepts and how to use sound level meters (SLMs).

This second tutorial focuses on discussing the rationale and the assumptions of MPANLs, how to measure ambient noise levels, how to read the tables with the MPANLs for different transducers, and how to adapt the MPANLs to test hearing thresholds in noisy environments and their ramifications. Most discussions are pertinent to the standards issued by ANSI/ASA and ISO, which are widely adopted around the world. Once one understands how to measure the ambient noise levels and apply the principles behind the MPANLs of one standard, they can apply the MPANL values from the standard(s) of their choice. Future tutorials will discuss how to calibrate air conduction and bone conduction transducers.

2. Rationale and assumptions of MPANLs

Standards on MPANLs assume that the ambient noise at or below the specified MPANLs produce a negligible masking effect on the test signals when the test signals are presented at their reference threshold sound pressure level (RETSPL) specified in standards that define normal hearing thresholds. For examples, ANSI/ASA S3.1:1999 (R2018) assumes the masking effects of the ambient noise in the environment is less than 2 dB if the pure tone test signal is presented at 0 dB HL as specified in the ANSI/ASA S3.6:2018 standard for audiometric testing. The RETSPL at a frequency is the sound pressure level needed for a specified type of transducer (e.g., supra-aural headphone or insert earphone) to be measured in a specified coupler or ear simulator (e.g., HA1 or HA2) so that the average sound pressure level at the ear drum of a large number of otologically normal listeners between the age of 18 and 30 years would yield a hearing threshold of 0 dB HL. An otologically normal listener is a person with normal state of health who does not have any signs or symptoms of ear diseases or obstructing ear wax, any history of undue exposure to noise or ototoxic drugs, or any hereditary hearing loss (IEC 60645–1, 2017).

Tables 1–3 tabulate the MPANLs specified by ANSI/ASA S3.1:1999 (R2018) for testing hearing thresholds from 125 to 8000 Hz, 250–8000 Hz, and 500–8000 Hz. Due to the limited space, the tables in ISO standards are not included. Notice in Tables 1–3 that the MPANLs at 125 and 250 Hz are different, depending on the intended hearing threshold testing range. The differences reflect the considerations for upward spread of masking, which occurs when low frequency sounds reduce the audibility of high frequency sounds (Egan & Hake, 1950). Physiologically, the peaks of the traveling waves for low frequency sounds (i.e., the characteristic frequency place) are closer to the apical end in the tonotopically organized basilar membrane and the peaks for the high frequency sounds are closer to the basal end. When the energy of a louder low frequency sound enters the cochlea, the traveling wave travels to and peaks at the apical end. This traveling wave also excites some neurons along the basilar membrane even in the absence of a high frequency sound. In order for a high frequency sound to be perceivable, it needs to generate a traveling wave peak that is higher than the traveling

Table 1
Maximum permissible ambient noise levels specified by ANSI/ASA S3.1:1999 (R2018) for testing hearing thresholds down to 0 dB HL from 125 Hz to 8000 Hz. Values are entered to illustrate how to determine whether the test room is appropriate for testing hearing thresholds down to 0 dB HL using different transducers.

ANSI/ASA S3.1:1999 (R2018) MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL) For Testing Thresholds from 125-8000 Hz						Center Frequency (Hz)	Measured Noise Level (dB SPL) ☐ 1 OB ☒ 1/3 OB	OK to Test Down to 0 dB HL?		
Ears Covered				Ear Uncovered						
ER 3A Insert Earphones		TDH Supra-Aural Headphones w MX-51 Cushion		Bone Conductor / Sound Field				ER Inserts	TDH Supra-aurals	BC / SF
1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band					
59.0	**54.0**	35.0	**30.0**	29.0	**24.0**	**125**	**35.2**	Yes	No	No
53.0	**48.0**	25.0	**20.0**	21.0	**16.0**	**250**	**18.4**	Yes	Yes	No
50.0	**45.0**	21.0	**16.0**	16.0	**11.0**	**500**	**10.8**	Yes	Yes	Yes
	44.0		**19.0**		**10.0**	**750**				
47.0	**42.0**	26.0	**21.0**	13.0	**8.0**	**1000**				
	43.0		**25.0**		**9.0**	**1500**				
49.0	**44.0**	34.0	**29.0**	14.0	**9.0**	**2000**				
	46.0		**33.0**		**8.0**	**3000**				
50.0	**45.0**	37.0	**32.0**	11.0	**6.0**	**4000**				
	48.0		**32.0**		**8.0**	**6000**				
56.0	**51.0**	37.0	**32.0**	14.0	**9.0**	**8000**				

Table 2
Table 2. Maximum permissible ambient noise levels specified by ANSI/ASA S3.1:1999 (R2018) for testing hearing thresholds down to 0 dB HL from 250 Hz to 8000 Hz. The MPANLs at 125 and 250 Hz are boxed to show the 14 dB/octave increase in MPANLs to account for upward spread of masking.

ANSI/ASA S3.1:1999 (R2018) MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL) For Testing Thresholds from 250-8000 Hz						Center Frequency (Hz)	Measured Noise Level (dB SPL) □ 1 OB □ 1/3 OB	OK to Test Down to 0 dB HL?		
Ears Covered				Ear Uncovered						
ER 3A Insert Earphones		TDH Supra-Aural Headphones w MX-51 Cushion		Bone Conductor / Sound Field				ER Inserts	TDH Supra-aurals	BC / SF
1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band					
67.0	**62.0**	39.0	**34.0**	35.0	**30.0**	**125**				
53.0	**48.0**	25.0	**30.0**	21.0	**16.0**	**250**				
50.0	**45.0**	21.0	**16.0**	16.0	**11.0**	**500**				
	44.0		**19.0**		**10.0**	**750**				
47.0	**42.0**	26.0	**21.0**	13.0	**8.0**	**1000**				
	43.0		**25.0**		**9.0**	**1500**				
49.0	**44.0**	34.0	**29.0**	14.0	**9.0**	**2000**				
	46.0		**33.0**		**8.0**	**3000**				
50.0	**45.0**	37.0	**32.0**	11.0	**6.0**	**4000**				
	48.0		**32.0**		**8.0**	**6000**				
56.0	**51.0**	37.0	**32.0**	14.0	**9.0**	**8000**				

Table 3
Table 3. Maximum permissible ambient noise levels specified by ANSI/ASA S3.1:1999 (R2018) for testing hearing thresholds down to 0 dB HL from 500 Hz to 8000 Hz. The MPANLs at 125, 250, and 500 Hz are boxed to show the 14 dB/octave increase in MPANLs to account for upward spread of masking.

ANSI/ASA S3.1:1999 (R2018) MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL) For Testing Thresholds from 500-8000 Hz						Center Frequency (Hz)	Measured Noise Level (dB SPL) □ 1 OB □ 1/3 OB	OK to Test Down to 0 dB HL?		
Ears Covered				Ear Uncovered						
ER 3A Insert Earphones		TDH Supra-Aural Headphones w MX-51 Cushion		Bone Conductor / Sound Field				ER Inserts	TDH Supra-aurals	BC / SF
1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band					
78.0	**73.0**	49.0	**44.0**	44.0	**39.0**	**125**				
64.0	**59.0**	35.0	**30.0**	30.0	**25.0**	**250**				
50.0	**45.0**	21.0	**16.0**	16.0	**11.0**	**500**				
	44.0		**19.0**		**10.0**	**750**				
47.0	**42.0**	26.0	**21.0**	13.0	**8.0**	**1000**				
	43.0		**25.0**		**9.0**	**1500**				
49.0	**44.0**	34.0	**29.0**	14.0	**9.0**	**2000**				
	46.0		**33.0**		**8.0**	**3000**				
50.0	**45.0**	37.0	**32.0**	11.0	**6.0**	**4000**				
	48.0		**32.0**		**8.0**	**6000**				
56.0	**51.0**	37.0	**32.0**	14.0	**9.0**	**8000**				

wave caused by the louder low frequency sound. The increase in level for a high frequency sound to overcome the presence of the low frequency sound is the amount of upward spread of masking.

Notice in Tables 1–3 that the MPANLs are transducer- and frequency-dependent. Both insert earphones and supra-aural headphones are considered ears covered conditions. As the foam tips of insert earphones (Etymotic Research) are made of hearing protection materials, they can greatly attenuate the noise level if the tips are inserted deep in the ear canal (i.e., the noise level inside the ear canal is much lower than the ambient noise level, Killion et al.,1985; Berger & Killion, 1989; Clark & Roeser, 1988). The MPANLs for insert earphones are, therefore, higher than those for the supra-aural headphones.

Sometimes, the MPANLs are referred to as "ears covered" versus "ears uncovered" conditions instead of the names of the transducers. For example, ear uncovered conditions are applicable to both bone-conduction or sound field hearing threshold tests because one or both ears of the listener is usually not covered by any earphones/headphones. Thus, the MPANLs for bone condition and sound field testing are the lowest (i.e., the most stringent).

In addition, MPANLs for ears covered condition was determined when both ears of the listener are wearing these transducers (i.e., both ears covered). Some old audiometers might only have one headphone/earphone (e.g., Starkey AA30) and the clinicians are supposed to put the headphone on whichever ear they want to test and the non-test ear is not covered (i.e., open). In such case, the ears uncovered MPANLs need to be used for determining the suitability of the room for audiometric testing.

Because of the differences in MPANLs, a testing environment may be suitable for testing hearing thresholds down to 0 dB HL using insert earphones or supra-aural headphones, but would not be for bone-conduction or sound field testing. As such, once the ambient noise levels are measured at different frequencies in a test room, one needs to determine the lowest testable threshold for each transducer.

The ANSI/ASA S3.1:1999 (R2018) standard assumes the upward spread of masking amounts to be at 14 dB per octave. In other word, the MPANL of the frequency that is one octave below the lowest test frequency is defined as 14 dB/octave higher than the MPANL of the lowest test frequency. For example, Table 2 states the MPANLs for testing between 250 and 8000 Hz. The MPANL requirement of 125 Hz which is 14 dB higher than the MPANL of 250 Hz which is the lowest test frequency. Similar requirement is also tabulated in Table 3 for testing between 500 and 8000 Hz. The MPANL of 125 Hz is 14 dB higher than that of 250 Hz, which is 14 dB higher than the lowest test frequency of 500 Hz. ISO 8253-1: (2010) (R2021) standard, on the other hand, assumes the 14 dB/octave pattern to 2 octaves below the lowest test frequency.

Further, standards assume that the measured levels are conducted under the worst acoustic conditions (ANSI/ASA S3.1:1999 (R2018)). It is important to measure the ambient noise level while testing patients or at the normal clinical testing environment when all the activities inside or surrounding the test room are in session (e.g., fans on, air-conditioning on, listeners waiting outside of the test room, other activities are carrying on outside and around the test room). Otherwise, the measurement would not reflect the normal ambient noise levels in the environment and may lead to erroneous conclusions about the suitability of the testing room for audiometric evaluations and the validity of the audiometric test results.

Both ANSI/ASA S3.1:1999 (R2018) and ISO 8253-1: (2010) (R2021) standards assume the ambient noise levels are measured by Class 1 SLMs (IEC 61672–1: 2013). Ambient noise levels should be measured annually in any clinic, any time when the acoustic environment in or around the test room changes, and whenever the audiometer is moved to a new location/room. The measurements are especially important for field testing in industrial and humanitarian service sites. Keeping the ambient noise levels below the MPANL specified in the standards ensures that listeners' hearing thresholds can be tested down to 0 dB HL in the test room.

3. Measuring the ambient noise levels using sound level meters (SLMs)

The equipment setup for measuring ambient noise levels is relatively simple but attention to details is very important. The following are the step-by-step procedures for conducting ambient noise measurements using Class 1 SLMs. Note that one needs to consult with the SLM manual because the procedures to adjust the SLM settings may differ.

1) Calibrate the SLM using a pistonphone or a sound calibrator and ensure the SLM reading matches the output level of the pistonphone/sound calibrator.
2) Set the parameters on the SLM to:
   - Microphone: a free-field microphone.
   - Range: the lowest setting (e.g., from 0 to 70 dB SPL) or a higher setting if the ambient noise level in any frequency band is higher than 70 dB SPL. The easiest way to check whether the measurement range is correct is to look at the display on the SLM. The top of the bar indicating the sound pressure level of the interested octave or 1/3 octave band should not exceed the top of the screen or fall below the bottom of the screen.
   - Frequency weighting: F (flat) or Z-weighting
   - Time weighting: S (slow)
   - Filtering: 1/3 octave band (preferred by ANSI/ASA S3.1:1999 (R2018)) or octave band if 1/3 octave band filters are not available.
3) Place the microphone of the SLM at the location of the listener's head in the test room. This location is usually at the center of the test room as regards to length and width, and at the height of the center of loudspeaker diaphragm.
4) Make sure no object is in the room to reduce or increase the ambient noise levels as in a normal testing session. If the person performing the calibration is in the testing room, make sure the person is not positioned to block a noise source or to create any extra noise in the room.
5) Determine the desired audiometric testing range and choose the applicable table (i.e., Tables 1, 2, or 3);
6) Switch the octave band or 1/3 octave band filter to 125 Hz.
7) Keep quiet and wait for the reading on the SLM to stabilize.
8) Read the sound pressure level measure in the octave band or 1/3 octave band. The screen of an SLM may display levels measured using multiple calculation methods. Be careful to read the correct level on the screen.
9) Record the ambient noise level in the choose table.
10) Increase the octave band or 1/3 octave band filter frequency.

11) Repeat Steps 7–10 till the measurements are made at all the frequencies.
12) Determine if the measured ambient noise level is lower than the MPANL for different transducers at the corresponding frequencies for the octave or 1/3 octave band used. If the measured level is lower than the specified MPANL for the transducer at a frequency, the test room is suitable for testing hearing thresholds down to 0 dB HL using the transducer at the test frequency.
13) Record the finding:
    a Yes – the ambient noise is low enough to use the transducer to test down to 0 dB HL in the test room
    b No – the ambient noise is too high to use the transducer to test down to 0 dB HL in the test room

Table 1 shows three examples of ambient noise levels measured by an SLM using 1/3 octave band filters and how to determine whether the test room can be used to test hearing thresholds down to 0 dB HL for different transducers. In the 125 Hz example, the measured level is 35.2 dB SPL which is lower than the 54.0 dB SPL permitted for insert earphones but higher than the 30.0 and 24.0 dB SPL permitted for supra-aural headphones and the ears uncovered conditions respectively. "Yes", "No", and "No" are, therefore, entered under insert earphones, supra-aural headphones, and BC/SF, respectively, to indicate it is possible to test down to 0 dB HL in the test room using insert earphones but not supra-aural headphones, bone conductors, or loudspeakers in sound field.

In the 250 Hz example, the measured ambient noise level is 18.4 dB which is lower than the MPANLs for both insert earphones (i.e., 48 dB SPL) and supra-aural headphones (20 dB SPL) but higher than that for ears uncovered conditions (16 dB SPL). The test room, therefore, is suitable for testing hearing thresholds down to 0 dB HL using insert earphones and supra-aural headphones but not for BC or sound field tests.

The 500 Hz example shows an example in which the measured ambient noise level (10.8 dB SPL) is lower than the MPALNs for all transducers. Three "Yes" are, therefore, entered to the table to show that the test room is appropriate to test hearing thresholds down to 0 dB HL for all the transducers. One needs to use the same procedures to determine whether the test area is suitable for assessing the hearing thresholds at all test frequencies for each transducer.

When using SLMs, one needs to keep in mind that they always display the combined/overall sound pressure level of the sounds from the environment and its self-generated noise. In a sound room located in a quiet area, the ambient noise level can be very low and it can be comparable to the self-noise levels of the SLMs (e.g., within 10 dB of the measured ambient noise level). ANSI/ASA S3.1:1999 (R2018) requires the self-noise of the SLM to be at least 3 dB below the measured ambient noise levels. Otherwise, one should seek a different SLM to measure the ambient noise level. On the other hand, if the measured level is 10 dB or more higher than the self-noise level, no additional correction is needed.

When self-noise level is between 3 and 10 dB of the measured level, however, ANSI/ASA S3.1:1999 requires the calculation of the actual ambient noise levels. This step is especially important if the measured level is slightly higher than the MPANL specified for a transducer under consideration. In such case, Formula 1 can be used to calculate the actual ambient noise level, which is used to determine the suitability of the test area for audiometric testing:

Table 4

Table 4. A table for recording the SLM self-noise levels, the measured ambient noise levels (which are overall levels of the SLM self-noise and the external noise in the background), and the actual ambient noise levels for the determination of the suitability of an area for audiometric testing. The maximum permissible ambient noise levels are specified by ANSI/ASA S3.1:1999 (R2018) for testing hearing thresholds down to 0 dB HL from 125 Hz to 8000 Hz.

MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL) For Testing Thresholds from 125-8000 Hz						Center Frequency (Hz)	SLM Self-Noise (dB SPL) □ 1 OB □ 1/3 OB	Measured Ambient Noise Level (dB SPL) □ 1 OB □ 1/3 OB	**Actual Ambient Noise Level (dB SPL) □ 1 OB □ 1/3 OB	OK to Test Down to 0 dB HL?		
Ears Covered				Ears Uncovered								
ER Insert Earphones		TDH Supra-Aural Headphones		Bone Conduction / Sound Field						ER Inserts	TDH Supra-aurals	BC / SF
1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band							
59.0	**54.0**	35.0	**30.0**	29.0	**24.0**	**125**						
53.0	**48.0**	25.0	**20.0**	21.0	**16.0**	**250**						
50.0	**45.0**	21.0	**16.0**	16.0	**11.0**	**500**						
	44.0		**19.0**		**10.0**	**750**						
47.0	**42.0**	26.0	**21.0**	13.0	**8.0**	**1000**						
	43.0		**25.0**		**9.0**	**1500**						
49.0	**44.0**	34.0	**29.0**	14.0	**9.0**	**2000**						
	46.0		**33.0**		**8.0**	**3000**						
50.0	**45.0**	37.0	**32.0**	11.0	**6.0**	**4000**						
	48.0		**32.0**		**8.0**	**6000**						
56.0	**51.0**	37.0	**32.0**	14.0	**9.0**	**8000**						

ANSI S3.1: 1999 (R2018) - Standard on Maximum Permissible Ambient Noise Levels
***Values based on Swanepoel and colleagues (2015)**
****Enter the Actual Ambient Noise level if the Measured Ambient Noise level is with 3-10 dB of the SLM Self-Noise levels.**

*Values based on Swanepoel et al. (2015)).
**Enter the Actual Ambient Noise level if the Measured Ambient Noise level is with 3–10 dB of the SLM Self-Noise levels.

$$L_{ActualAmN} = 10\log\left(10^{LMeasuredAmN/10} - 10^{LSLMSelfN/10}\right),\ \text{dB SPL} \quad (1)$$

If self-noise needs to be calculated, Table 4 can be used to record the sound pressure levels. After filling in the self-noise at different levels and the measured levels, one needs to make decision about the suitability of the test area for audiometric testing using the column showing the actual ambient noise levels. For example, the MPANL at 1000 Hz is 8 dB SPL for the ears uncovered condition. The measured ambient noise level is 10 dB SPL, which results in the conclusion that bone conduction thresholds cannot be tested down to 0 dB HL in the room. Upon close inspection, however, self-noise of the SLM was measured to be 6 dB SPL at 1000 Hz when the microphone is covered by an insulation cap. As the SLM self-noise is between 3 and 10 dB of the measured ambient noise level. Formula 1 is used to calculate the actual ambient noise level:

$$L_{ActualAmN} = 10\log\left(10^{10/10} - 10^{6/10}\right),\ \text{dB SPL}$$

$$L_{ActualAmN} = 7.8\,\text{dB SPL}$$

This shows that the actual ambient noise level was lower than 8 dB SPL. The test room, therefore, is suitable for testing hearing thresholds down to 0 dB HL.

4. What to do when the ambient noise level is too high in a test room

If the measured levels are higher than the MPANLs stated in the standards for testing hearing thresholds down to 0 dB HL, interventions should be made to either lower the ambient noise levels in and around the test room or to use transducers with higher MPANLs. Such intervention efforts can grossly be divided to several categories:

a ***Modify the physical environment***, e.g.,
- (if a sound booth is newly installed) Contact the manufacturer of the sound booth to resolve possible installation problems.
- Acquire a sound booth with higher attenuation properties.
- Add absorbent materials to the walls, ceiling, or floor inside and outside of the test room and make sure seal is obtained as sounds can travel through the cracks.
- Replace noisy fans and oil squeaky doors.
- Turn off the air conditioning and close the windows while conducting hearing tests/screenings.

b ***Move the test room to a quieter location***, e.g.,
- Measure the ambient noise in several alternative locations and then choose the space with ambient noise levels below the MPANLs.

c ***Remove the noise sources in and around the test room***, e.g.,
- Move the waiting room/space to be far away from the test room.
- Move the noise-generating activities/equipment to outside of the test room or to location(s) far away from the test room; or

d ***Make administrative adjustment***, e.g.,
- Schedule and conduct hearing tests at a time/on the days that activities in and around the test room are the minimum.

e ***Use transducers with higher MPANLs***, e.g.,
- Invest in transducers with higher attenuation properties (i.e., insert earphones, circumaural headphones, or KuduWave audiometers, Fig. 1).

f ***Increase the lowest threshold level can be assessed in the room,*** e.g.,
- Raise the lowest threshold level can be tested in the room. If one decides to test hearing thresholds down to 20 dB HL instead of to 0 dB HL, the MPANLs can be raised by 20 dB).

g ***Reduce the frequency range of the hearing test,*** e.g.,
- Raise the frequency range to a higher frequency. Instead of testing from 250 to 4000 Hz, one can test from 500 to 4000 Hz to raise the MPANLs at low frequencies (see the differences in MPANLs in Tables 1–3).



Fig. 1. A. KuduWave components. B. KuduWave worn on a child for testing hearing thresholds in the field.

Readers interested in measures to reduce room noise are encouraged to seek professional help or guidance from books in room acoustics. As the strategies outlined in a) to d) are either self-explanatory or out of the scope of audiometric calibration, the following discussion focuses on the calibration concepts and applications related to strategies e) to g).

4.1. Use transducers with higher MPANLs

Insert earphones typically provide higher noise attenuation than supraural headphones when they are inserted deep in the ear canal (Killion et al., 1985; Berger & Killion, 1989; Clark & Roeser, 1988). A caution is that clinicians need to make sure the ear canals do not have wax accumulation before insertion. Otherwise, the inserts can push the ear wax further into the ear canal. In such case, circumaural headphones can be an alternative in rooms that are not quiet enough for supra-aural headphones but sufficiently quiet for their use.

Margolis and colleagues conducted a series of studies to examine the amount of attenuation provided by circumaural headphones (Margolis et al., 2022; Smull et al., 2019). They concluded that circumaural headphones generally provided more attenuation to ambient noise than supra-aural headphones and they subsequently derived the MPANLs for different types of circumaural headphones (Table 5). The advantage of circumaural headphones is that, like supra-aural headphones and insert earphones, they can be used by any audiometer after being properly calibrated.

Another option is KuduWave, which is a computer-based audiometer that uses insert earphones and an additional circumaural earcup to reduce background noise (Fig. 1). Air conduction (AC) thresholds are tested using the insert earphones and BC thresholds are tested with a bone conductor placed on the listener's forehead. A pair of circumaural earcups is worn during the testing of both types of thresholds. The MPANLs for the two transducers are, therefore, identical for both tests (i.e., ears covered). As both circumaural earcups and insert foam tips provide extra noise attenuation, the MPANLs, in theory, should be higher than insert earphones alone.

Although the MPANLs for the KuduWave audiometers have not been incorporated into any national or international standards, Swanepoel et al. (2015) published a set of MPANLs in octave bands based on the data obtained in an empirical study (Table 5). Their results show that KuduWave provided 1 – 10 dB of additional ambient noise attenuation at various frequencies compared to the insert earphones. Stangl (2015) also reported similar results.

4.2. Increase the lowest threshold level that can be assessed in the room

Hearing screening is often carried out in field testing environments with higher ambient noise levels than sound rooms. As the goal is to separate listeners with normal hearing from those with hearing loss in a hearing screening, the passing criteria is often set to a higher hearing level (e.g., 20 dB HL). Remember that the MPANLs specified in ANSI/ASA S3.1:1999 (R2018) and the ISO 8253-1:(2010) (R2021) standards are used for testing hearing thresholds down to 0 dB HL. As Hawkins and Stevens (1950) showed that every 1 dB increase in masking noise shifted listeners' hearing threshold by 1 dB, the converse is true that setting the lowest hearing level to 20 dB HL would allow 20 dB to be added to the MPANLs stated in Tables 1–5.

Table 6 shows a new MPANL table for testing hearing thresholds down to 20 dB HL. The rationale is that if the lowest test signal level is 20 dB HL, the ambient noise can be 20 dB higher and its masking effect is still negligible for the test signal. In other words, the

Table 5

Table 5. The MPANLs of circumaural headphones obtained empirically by Margolis and colleagues (2022) MPANLs of three types of circumaural headphones. The MPANLs of insert earphones specified in ANSI/ASA S3.1:1999 (R2018) are included as references.

MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL)									Center Frequency (Hz)	Measured Noise Level (dB SPL) □ 1 OB □ 1/3 OB	OK to Test Down to 0 dB HL?	
For testing hearing thresholds:												
250-8000 Hz						125-8000 Hz						
Ears Covered												
Circumaural Earphones*						Insert Earphones AC		KuduWave** AC & BC				
Radio Ear DD450		Sennheiser HD280Pro		Radio Ear DD65v2							ER Inserts	KuduWave AC & BC
1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band				
48.0	**43.0**	32.0	**27.0**	41.0	**36.0**	59.0	**54.0**	67.0	**125**			
37.0	**32.0**	21.0	**16.0**	37.0	**32.0**	53.0	**48.0**	59.0	**250**			
39.0	**34.0**	28.0	**23.0**	42.0	**37.0**	50.0	**45.0**	60.0	**500**			
42.0	**37.0**	29.0	**24.0**	43.0	**38.0**		**44.0**		**750**			
43.0	**38.0**	30.0	**25.0**	46.0	**41.0**	47.0	**42.0**	54.0	**1000**			
45.0	**40.0**	36.0	**31.0**	48.0	**43.0**		**43.0**		**1500**			
47.0	**42.0**	41.0	**36.0**	58.0	**53.0**	49.0	**44.0**	51.0	**2000**			
52.0	**47.0**	38.0	**33.0**	52.0	**47.0**		**46.0**		**3000**			
58.0	**53.0**	39.0	**34.0**	55.0	**50.0**	50.0	**45.0**	59.0	**4000**			
58.0	**53.0**	36.0	**31.0**	59.0	**54.0**		**48.0**		**6000**			
58.0	**53.0**	32.0	**27.0**	60.0	**55.0**	56.0	**51.0**	57.0	**8000**			

***Values from Margolis and colleagues (2022)**
****Values from Swanepoel and colleagues (2015)**

*Values from Margolis et al. (2022)).
**Values from Swanepoel et al. (2015)).

listener can withstand a higher amount of ambient noise and still be able to hear the test stimuli. This increase in MPANLs can be welcomed or needed in field testing environments where noise level is higher than those in quiet sound booths.

By the same logic, if clinicians want to test listeners' hearing thresholds below a calibrated 0 dB HL (e.g., down to −10 dB HL), the ANSI/ASA S3.1:1999 (R2018) recommends that the ambient noise levels of the test room should be 10 dB less than the MPANLs specified in the standards to ensure the test room is sufficiently quiet to allow accurate testing of hearing thresholds down to −10 dB HL.

Changing the lowest testable hearing thresholds (or increasing MPANLs) in a hearing screening is easy to implement and the results are easy to interpret if only air conduction (AC) thresholds are tested. Changing the lowest testable hearing thresholds in a clinical setting, however, should be avoid because the interpretation of test results can be complicated and is prone to making mistakes. If the change has to occur, it is important to convey this room limitation to everyone who would test hearing thresholds and who would receive and interpret the test results. The following describes scenarios in which changing the lowest testable hearing thresholds becomes complicated.

Scenario 1: As the different transducers have different MPANLs (Tables 1–4), some transducers may allow testing hearing thresholds down to 0 dB HL (e.g., insert earphones) and others may only allow testing down to 20 dB HL (e.g., bone conductors). Clinicians need to pay special attention when interpreting listeners' hearing thresholds in this range. Specifically, a listener with normal hearing (e.g., AC and BC thresholds = 0 dB HL) can have air conduction (AC) thresholds that are 20 dB lower/better than the BC thresholds. This phenomenon is understandable and logical once the room acoustics are taken into account. But it might be treated as an oddity if the clinician or the interpreter does not know the limitations of the test room.

Scenario 2: in the same room that can test AC thresholds down to 0 dB HL using insert earphones but BC thresholds to 20 dB HL, the clinician has limited ability to identify 1) listeners having a mild/moderate conductive/mixed hearing loss or 2) listeners having normal hearing but with a conductive component. To illustrate the problem, assume that a listener's actual AC thresholds = 30 dB HL and BC thresholds = 0 dB HL. With the room limitations, the listener would have AC thresholds = 30 dB HL and BC thresholds = 20 dB HL. The air-bone gap of 10 dB would be considered as an acceptable air-bone gap and, therefore, the conductive hearing loss would be missed.

These two examples highlight the importance of informing all people and professionals involved in the hearing testing process and those who receive and interpret the audiograms, the limitations of the test room and its ramifications. They also emphasize the importance of having alternative testing equipment (e.g., tympanometry or reflectance tests to inform listeners' outer and middle ear status or otoacoustic missions or (automatic) auditory brainstem responses to inform listeners' hearing sensitivity).

As some noises have energy concentrated in some frequency regions, one may be tempted to determine the lowest hearing threshold assessable in a test area frequency-by-frequency instead of down to a fixed dB HL at all frequencies. This approach is theoretical feasible but practically complex and it is not recommended in ANSI/ASA standard. At high frequencies, the application is straightforward. For example, in the summer, audiometric testing may need to be carried out in the presence of cicada songs which have the most prominent frequency at ~4000 Hz (Bennet-Clark, 1998). If the measured sound pressure level of the cicada songs is 58.6 dB at 4000 Hz (Table 7). One can calculate how much the cicada songs are higher then the MPANLs at 4000 Hz for different transducers

Table 6
Table 6. The MPANLs for testing hearing thresholds down to 20 dB HL at frequencies 125-8000 Hz.

MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL) For Testing Thresholds from 125-8000 Hz							Center Frequency (Hz)	Measured Noise Level (dB SPL) □ 1 OB □ 1/3 OB	OK to Test Down to 20 dB HL?		
Ears Covered					Ears Uncovered						
KuduWave*	ER Insert Earphones		TDH Supra-Aural Headphones		Bone Conduction / Sound Field				ER Inserts	TDH Supra-aurals	BC / SF
1 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band	1 Octave Band	1/3 Octave Band					
87.0	79.0	**74.0**	55.0	**50.0**	49.0	**44.0**	**125**				
79.0	73.0	**68.0**	45.0	**40.0**	41.0	**36.0**	**250**				
80.0	70.0	**65.0**	41.0	**36.0**	36.0	**31.0**	**500**				
		64.0		**39.0**		**30.0**	**750**				
74.0	67.0	**62.0**	46.0	**41.0**	33.0	**28.0**	**1000**				
		63.0		**45.0**		**29.0**	**1500**				
71.0	69.0	**64.0**	54.0	**49.0**	34.0	**29.0**	**2000**				
		66.0		**53.0**		**28.0**	**3000**				
79.0	70.0	**65.0**	57.0	**52.0**	31.0	**26.0**	**4000**				
		68.0		**52.0**		**28.0**	**6000**				
77.0	76.0	**71.0**	57.0	**52.0**	34.0	**29.0**	**8000**				

ANSI S3.1: 1999 (R2018) - Standard on Maximum Permissible Ambient Noise Levels (Reaffirmed in 2018)
***Values based on Swanepoel and colleagues (2015)**
****Enter the Actual Ambient Noise level if the Measured Ambient Noise level is with 3-10 dB of the SLM Self-Noise levels.**

*Values based on Swanepoel et al. (2015)).
**Enter the Actual Ambient Noise level if the Measured Ambient Noise level is with 3–10 dB of the SLM Self-Noise levels.

(i.e., 58.6–45.0 = 13.6 dB for insert earphones, 58.6–32.0 = 26.6 dB for supra-aural headphones, and 58.6–6.0 = 52.5 dB). The minimum hearing thresholds can be assessed in the test area are, therefore, 15, 30, and 55 dB HL for insert earphones, supra-aural headphones, and bone vibrators/loudspeakers, respectively, after rounding up to the nearest multiples of 5 dB.

For low frequencies, however, the dual masking effects by upward spread of masking and by background noise with sound components surrounding the test frequency (i.e., on-frequency masking) need to be considered. Consider the scenario in Table 1, the measured noise level are 35.2, 18.4, and 10.8 dB SPL at 125, 250, and 500 Hz, respectively (Table 7). The calculation of the on-frequency masking effect is straightforward and similar to the procedures at 4000 Hz.

When the upward spread of masking is considered from 1 octave band below, the 35.2 dB SPL of ambient noise at 125 Hz has a 14 dB per octave of masking effect on test stimuli at higher frequencies. In other words, the masking effect of the ambient noise at 125 Hz is weakened by 14 dB at 250 Hz. This means that if 35.2 dB SPL minus the MPANL at 250 Hz for a transducer is less than 14 dB, the ambient noise at 125 Hz has no masking effect on the test signal. Otherwise, the masking effect of the ambient noise at 125 Hz needs to be considered when determining the lowest hearing threshold assessable at 250 Hz for different transducers. In this particular example, the masking noise at 125 Hz (i.e., 35.2 dB SPL) minus the MPANL at 250 (i.e., 48 dB SPL) minus 14 dB of reduction in upward spread of masking is less than 0 dB for insert earphones, thus no further masking effect is created by the ambient noise at 125 Hz.

For other transducers, however, 35.2 – 20.0 – 14.0 = 1.2 dB for supra-aural headphones and 35.2 – 16.0 – 14.0 = 5.2 dB in ears uncovered conditions. When these numbers are rounded up to the nearest multiples of 5, upward spread of masking effect at 250 Hz becomes 5 and 10 dB HL for supra-aural headphones and ears uncovered conditions, respectively. As the masking on test stimuli is created by the louder masker, the lowest hearing thresholds assessable at 250 Hz become 5 dB HL for supra-aural headphones and 10 dB HL for ears uncovered conditions after both the on-frequency and the upward spread of masking effects are taken into account.

This example illustrates the complexity of determining frequency-dependent lowest hearing thresholds assessable in a test room. In some cases, the low frequency energy is so high that its influence goes beyond one octave up above. In such case, the upward spread of masking effects need to be considered several octaves above the frequency of the ambient noise and the calculations of the lowest hearing threshold assessable in a test room become very complex.

A simpler method is to test the hearing thresholds of two otologically normal listeners between 18 and 25 years old. As the testing condition has already accounted for all the masking effects, the hearing thresholds of the two listeners should reflect the lowest hearing thresholds assessable in the test area. If test area has fluctuating noise, however, the benefits of this psychoacoustic calibration procedure is nullified (see more details below in 3.4). In such case, KuduWave might be the best option because it has the highest MPANLs among the transducers for both ears covered (i.e., air condition) and ears uncovered conditions (i.e., bone conduction, Table 5).

4.3. Reduce the frequency range of the hearing test

When the ambient noise level of a test room exceeds the MPANLs at any frequency, the general recommendation is to disqualify the room for audiometric testing (ANSI/ASA S3.1:1999). An exception to the rule is when the test frequency range of audiometry is limited (e.g., from 1000 to 8000 Hz only). The ANSI/ASA standard suggests that if the ambient noise level is equal to or below the MPANLs at an octave frequency below the test frequency (i.e., 500 Hz in this case), it is possible to conduct hearing tests in the room. The specific example given in ANSI/ASA S3.1: 1999 (R2018) was: if the hearing screening is carried out at 20 dB HL using supra-aural headphones at 1000, 2000, and 4000 Hz, the MPANLs in the test room shall not exceed the levels shown in Table 6 for supra-aural headphones in the frequency range of 500 to 8000 Hz.

This exception, however, should be applied with caution. One of the reasons is that, due upward spread of masking in the auditory system, low frequency ambient noise can potentially have a higher amount of masking effect on high frequency pure tone stimuli than the ambient noise level measured at a higher frequency. For example, when the environment has excessive low frequency noise (e.g., 67 dB at 250 Hz measured using a 1/3 octave band filter) but the ambient noise level is equal to or less than the MPANLs for the

Table 7

Table 7. Examples at 125, 250, 500, and 4000 Hz are given to show the considerations in on-frequency masking and upward spread of masking when one attempts to determine the lowest frequency assessable at each frequency in a test area.

MAXIMUM PERMISSIBLE AMBIENT NOISE LEVEL (dB SPL)			Center Frequency (Hz)	Measured Noise Level (dB SPL)	On-Frequency Masking Effect						Upward Spread of Masking Effect						OK to Test Down to 0 dB HL?		
For Testing Thresholds from 125-8000 Hz																			
Ears Covered		Ears Uncovered			ER Inserts		TDH Supra-aurals		BC / SF		ER Inserts		TDH Supra-aurals		BC / SF		ER Inserts	TDH Supra-aurals	BC / SF
ER 3A Insert Earphones	TDH Supra-Aural Headphones w MX-51 Cushion	Bone Conductor / Sound Field		□ 1 OB ☒ 1/3 OB															
1/3 Octave Band	1/3 Octave Band	1/3 Octave Band			Dev Insert	Down to?	Dev TDH	Down to?	Dev BC/SF	Down to?	Dev Insert	Down to?	Dev TDH	Down to?	Dev BC/SF	Down to?			
54.0	30.0	24.0	125	35.2	<0	0	+5.2	10	11.2	15	-	-	-	-	-	-	0	10	15
48.0	20.0	16.0	250	18.4	<0	0	<0	0	2.4	5	<0	0	1.2	5	5.2	10	0	5	10
45.0	16.0	11.0	500	10.8	<0	0	<0	0	<0	0	<0	0	<0	0	<0	0	0	0	0
44.0	19.0	10.0	750																
42.0	21.0	8.0	1000																
43.0	25.0	9.0	1500																
44.0	29.0	9.0	2000																
46.0	33.0	8.0	3000																
45.0	32.0	6.0	4000	58.6	13.6	15	26.6	30	52.6	55							15	30	55
48.0	32.0	8.0	6000																
51.0	32.0	9.0	8000																

transducer at higher frequencies (e.g., supra-aural headphones, 18 dB at 1000 Hz). As the excessive low frequency noise can affect the audibility of high frequency test signals due to upward spread of masking in the auditory system, the effective masking by the low frequency ambient noise for the 1000 Hz tone would be higher than the 18 dB SPL as measured by the SLM. Using the 14 dB per octave rule, the effective masking effect of the ambient noise at 250 Hz is 67–28=39 dB SPL at 1000 Hz, in theory. As such kinds of occurrences may complicate the interpretations of test results, clinicians are encouraged to test two otologically normal listeners and check if all the test tones are audible at 0 dB HL. If not, consider increasing the lowest testable hearing threshold or disqualifying the test room for audiometric testing.

5. Biological/Psychoacoustic check of ambient noise

Measuring the ambient noise levels using SLMs is the gold standard for ensuring the test room are suitable for testing hearing sensitivity down to a 0 dB HL or an alternative hearing level. In case when an SLM is not available (e.g., in humanitarian service or other field-testing settings), ANSI/ASA S3.1:1999 (2018) recommends a biological/psychoacoustic calibration by at least two listeners with normal hearing. Listeners with normal hearing is defined as listeners with air conduction thresholds $\leq$0 dB HL from 125 – 8000 Hz in each ear when tested in a room having ambient noise levels equal to or lower than the MPANLS specified in the ANSI/ASA S3.1 standard.

The psychoacoustic calibration was originally named as biological calibration. The ANSI/ASA favors the term psychoacoustic calibration, likely because it refers to the ability of a listener to detect test stimuli in background noise. As the task involves the interactions between acoustic stimuli and human perception, it is a psychoacoustic task.

ANSI/ASA S3.1:1999 (R2018) stipulates that the audiometer needs to have already been calibrated to the RETSPLs specified in the ANSI/ASA S3.6 or ISO 389 standards *prior* to the psychoacoustic calibration. If the audiometer has been calibrated with correction factors, the audiometer dial setting needs to be changed so that the transducers always present the stimuli at a level equivalent to 0 dB HL. The assumption is that if the audiometer is functioning normally and if a person with normal hearing can detect the test stimuli in the ambient noise, the test area is quiet enough for others to detect the stimuli in the same condition.

In addition, the ambient noise should represent the worst possible noise conditions for audiometric testing in the test room at the time of the psychoacoustic calibration. All possible noises that can occur during audiometric testing should be present during the psychoacoustic calibration. These noises may include, but not limited to, fan noise, other equipment noise in the test room, noise from the waiting room, and noises from the surrounding rooms or outside of the window.

During the psychoacoustic calibration, pure tones are presented through different transducers at 0 dB HL and the two listeners with normal hearing need to hear each pure tone stimuli at least 2 times at each frequency. To perform the psychoacoustic calibration, a listener with normal hearing:

a Is placed in the normal testing location in the test room.
b Is tested at a time that the hearing tests are normally conducted.
c Responds to tones presented through a bone conductor or a loudspeaker at 0 dB HL at least two times at each test frequency.
d Listens to beats or any external noise that might prevent them from hearing the testing signal.

If two listeners could hear the pure tones/warble tones presented through the bone conductor and the loudspeaker at all the test frequencies for at least two times, the test room is deemed to be suitable for audiometric testing in both ears covered and ears not covered conditions. Otherwise, one would need to repeat the above procedures with a pair of supra-aural or insert earphones that are worn by the normal hearing listeners in a standard manner to determine the suitability of the test room for the individual ears covered conditions.

Besides determining the suitability of a test room for audiometric testing, the psychoacoustic calibration is often performed every day in clinics before seeing patients to ensure the equipment is functioning normally. This daily psychoacoustic check, however, is somewhat less formal because it is normally conducted by one person who opens the clinic to ensure the audiometer is functioning normally and free of statics or other abnormal noises.

6. Documenting ambient noise levels in field-testing sites

The ambient noise levels in clinics with sound booths are generally steady and low. In field-test settings, however, the ambient noise levels can fluctuate considerably or vary depending on the time of the day, making the ambient noise level estimation difficult. The ANSI/ASA S3.1:1999 (R2018) standard recommends measuring and recording the highest ambient noise levels in the test room when examining its suitability for audiometric testing and when determining the lowest testable hearing thresholds.

One of the ways to stabilize the SLM readings in such environment is to use the slow time-weighting to make the ambient noise measurement. ANSI/ASA S3.1:1999 (R2018) does not stipulate which time-weighting to measure ambient noise levels but the slow time-weighting generally gives a more stable reading than the fast time-weighting. The slow time-weighting, therefore, is generally a better setting for measuring ambient noises.

In some humanitarian service settings, however, a stabilized SLM reading may not be possible even with slow time-weightings because double-walled sound booths are rare and noise can come from many sources (e.g., people talking or walking, traffic noise, music, fans, air-conditioning noise coming from inside and outside of the testing room). A natural tendency of clinicians working in

these settings is to wait for the gaps in background noise and present the test stimuli during the relatively quieter periods of the noise. They may also present the test stimuli again if the stimuli were presented when the background noise was loud. Nevertheless, the presence of these noises makes it impossible to use a single number to reflect the complexity of the noise in the test room.

Although not recommended in any international standards, recording a noise profile may be one of the ways to resolve the problem of having many fluctuating noise in the environment. One can record multiple noise levels to document the minimum, maximum, average, 10th, 25th, 50th, 75th, and 90th percentile levels and the standard deviations of the noise fluctuations during normal clinical testing time.

The measured ambient noise levels cannot be used to compare with the MPANLs in ANSI/ASA or ISO standards to determine the suitability of the test room for audiometric testing, but they can create a noise profile and serve as a reference for clinical decisions and interpretations. Measurements such as the minimum, maximum, average levels, the standard deviations, and the 10th, 25th, 50th, 75th, and 90th percentile information can shed light on amount of noise fluctuations, the best, average, and worst acoustic conditions in the test room. For example, if the 90 percentile of ambient noise levels are approximately 30 dB higher than the MPANLs, the clinicians can consider several options:

- Change the test room.
- Employ measures to reduce background noise.
- Test listeners using audiometers calibrated with insert earphones or KuduWave to take advantage of their higher MPANLs.
- Raise the lowest testable hearing thresholds to be 30 dB HL.
- Use other tests to confirm or rule-out certain ear and hearing disorders, e.g., OAEs for outer hair cell integrity, tympanometry or reflectance measurements for outer and middle ear disorders.

If the strategy of waiting to present test stimuli during noise gaps allows clinicians to test thresholds around and below 30 dB HL, clinicians need to exert caution when interpreting audiograms with mild hearing loss as the listener's hearing sensitivity can be better. On the other hand, the clinicians can be assured that hearing thresholds higher than moderate hearing loss are accurate as the ambient noise is less likely to mask the test stimuli at those stimulus presentation levels.

As ambient noise levels are measured in octave and 1/3 octave bands, making multiple measurements at one frequency band at a time can be exceedingly time consuming. Fortunately, many SLMs have value reporting screens which displays the values measured in multiple octave or 1/3 octave bands simultaneously (Fig. 2). Some SLMs can be connected to a computer and save the measured levels in multiple bands as a file. Both options can considerably speed up the measurement process.

To document the variations of ambient noise levels in an environment with fluctuating ambient noise, one can:

1) Calibrate the SLM using a pistonphone or a sound calibrator and ensure the SLM reading matches the output of the pistonphone/sound calibrator.
2) Place the SLM at the location of the audiometry testing location.
3) Set the SLM to:
   - Range: the lowest setting (e.g., from 0 to 70 dB SPL) or a higher setting if the ambient noise level in any frequency band is higher than 70 dB SPL. The easiest way to check whether the measurement range is correct is to look at the display on the SLM. The bar indicating the sound pressure level of the interested octave or 1/3 octave band should not exceed the top of the screen.
   - Frequency weighting: F (flat) or Z-weighting
   - Time weighting: S (slow)

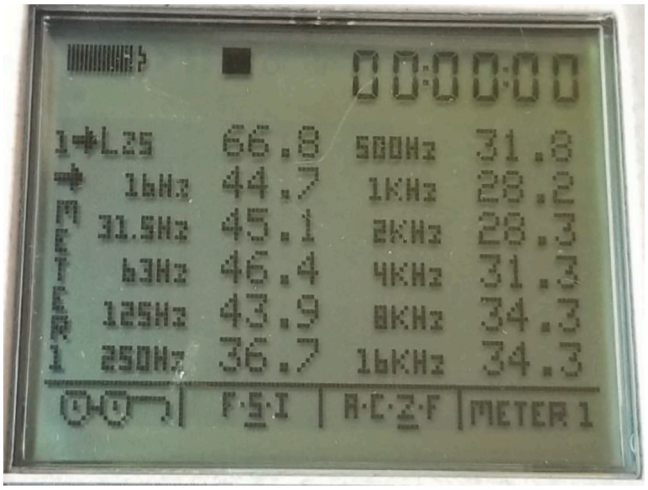


Fig. 2. Sound level meter screens showing sound pressure levels measured in octave bands with center frequencies from 16 Hz to 16,000 Hz.

- Filtering: 1/3 octave band, if available. Otherwise, octave band
- Multiple octave or 1/3 octave filter band screen displaying the values in each band OR Computer interface on

4) Set the timer on a stop-watch on a phone or a computer to 30 s;
5) Take a picture of the SLM display screen at the 30th second, OR

save the SLM readings into a file at the 30th second;

1) Change the frequency range of the octave or 1/3 octave band filter;
2) Repeat Steps 4–6 till all the frequencies from 125 to 8000 Hz are measured;
3) Repeat Steps 4–7 till 10 measurements of the whole frequency range are made.
4) Enter the values at the octave or 1/3 octave bands onto Table 8, OR

extract the data and arranged them into Table 8;

1) Enter the MPANLs for the appropriate test frequency range and filter setting in the Table 8;
2) Calculate the minimum, 10th, 25th, 50th, 75th and 90th percentiles, maximum, and average ambient noise levels at each frequency band. Note that the 50th percentile value is the median of the measurements which can be different from the average level which is the mean. (Table 8 with all the formulae to automatically calculate the percentile and average values can be downloaded here)
3) Determine whether the ambient noise is too loud for testing down to 0 dB HL at each frequency for each transducer.

Making multiple measurements at a constant time interval is important (e.g., every 30 s for 10 measurements) because the timer chooses which moments to take the measurements instead of a person. This practice reduces human bias in the measurement process.

If audiometric testing is conducted in multiple test rooms, the above measurements should be conducted in all the rooms. Regardless of the testing protocol, it is a good practice to measure and document the ambient noise levels in the field-testing room/environment because they may help the interpretation of the test results and clinical decisions later.

Additionally, noise in field-testing environment often change depending on the time of the day as the activities in the surrounding areas change. For example, if all the children being tested had hearing thresholds at 25 dB HL at 500 Hz when tested after lunch, the noise measurements show an elevated average ambient noise level at 125 Hz and 250 Hz (i.e., > the values shown in Table 2 plus 25 dB at these frequencies). Along with normal tympanometry/reflectance results, the test results can be interpreted as a result of elevated ambient noise levels and no further testing is required. On the other hand, if the same test results are obtained in a test environment with the measured ambient noise lower than 25 dB HL plus the values stated in Table 2, then the test results could hint an equipment failure.

Note that ANSI/ASA S3.1:1999 (R2018) recommends using the maximum ambient noise level to be compared with the MPANLs. It is essential to follow this recommendation if automatic hearing test applications/programs are used to test listeners because most automatic test program would present the stimuli without considering the instantaneous levels of the ambient noise levels. Although programs monitor the ambient noise levels (e.g., the automatic testing option of KuduWave by eMoyo), they do not consider the ambient noise levels or wait for noise gaps to present test stimuli. Clinicians, therefore, need to use the maximum ambient noise levels to determine the suitability of the room for audiometric testing and to interpretate the test results accordingly.

Caution should be exercised when testing the hearing thresholds of young children in noisy environments because they often respond at a level with higher than their thresholds (i.e., minimal response levels). It is likely that they would need a higher signal-to-noise ratio (i.e., ambient noise level needs to be lower) to respond to test stimuli. Clinicians testing in the field, therefore, need to consider the effects of background noise and seek alternative tests to examine or cross-check their hearing sensitivity or to verify the pure tone thresholds in quieter environments. Further, research on “distraction masking” indicated that the hearing thresholds of infants and adults increased when other sounds were present in the testing environments, even though the sounds did not create on-frequency masking (Margolis et al., 2022; Werner & Bargones, 1991). The amount of threshold shift is unpredictable and depends on the types of noise. Although the shift is generally less than the 1 dB threshold shift per 1 dB increase in noise level as observed in on-frequency masking, clinicians need to be mindful of the background noises in the testing environment and reduce distracting noise as much as possible.

7. Summary

Ambient noise measurements need to be conducted before testing hearing thresholds, especially in field testing locations where there are no sound rooms. SLMs are common and convenient tools to measure the ambient noise levels of test rooms. Ambient noise measurements are important for ensuring test rooms meet standards for audiometric testing, especially for people with normal hearing.

The MPANLs were recommended based on several factors and assumptions:

- The levels are measured in octave or 1/3 octave bands from 125 to 8000 Hz for conventional audiometry levels, or to 16,000 Hz for high-frequency audiometry.
- The ambient noise would only contribute to a negligible amount of masking effect (< 2 dB) to the test signals presented at the reference equivalent threshold sound pressure levels (RETSPLs) for 0 dB HL as specified in ANSI/ASA S3.6:2018 Specification of Audiometers for different transducers.

Table 8
A table for documenting the ambient noise variations in a test room.

FLUCTUATING AMBIENT NOISE LEVEL MEASUREMENT																					
Measurements in □ 1 OB □ 1/3 OB										Min Ambient Noise Level	Percentile					Max Permissible Ambient Noise Level	Standard Deviations	Average Ambient Noise Level	Center Frequency	MPANLs for the Transducer □ 1 OB □ 1/3 OB	OK to Test Down to 0 dB HL?		
1	2	3	4	5	6	7	8	9	10		10	25	50	75	90						ER Inserts	TDH Supra-aurals	BC / SF
																			125				
																			250				
																			500				
																			750				
																			1000				
																			1500				
																			2000				
																			3000				
																			4000				
																			6000				
																			8000				

- The attenuation provided by the cushion of supra-aural headphones or by the foam tip of the insert earphones can reduce the amount of ambient noise at the ear drum which may mask the test stimuli.
- Both ears are covered when wearing supra-aural or insert earphones, otherwise, ears uncovered permissible levels apply.
- Low-frequency ambient noise can mask high-frequency test signals due to upward spread of masking.
- When the background noise level fluctuates, the worst acoustic environment is measured.
- Self-noise of the SLM is at least 3 dB lower than the MPANL.
- Calculate the actual ambient noise level if the SLM self-noise level is between 3 and 10 dB of the measured ambient noise level which is a combination of the SLM self-noise and external ambient noise.

When a test room is too noisy for testing hearing thresholds down to 0 dB for a transducer, one can determine alternative threshold testing criteria based on the purposes of the testing. The MPANLs can be raised if one decides to raise the lowest threshold they want to test and reduce the range of test frequencies. Alternatively, one can employ measures to reduce the room noise levels by modifying the physical environment, moving the test room to a quieter location, removing the noise sources in and around the test room, adjusting the clinic schedule to a quieter time, and using transducers with higher MPANLs.

Disclaimer: The author is not associated with any company that manufactures or sells the calibration equipment and has no conflict of interest. The author has obtained written permission from eMoyo to use the pictures in Fig. 1.

References

ANSI/ASA S3.1:1999(R2018). Maximum permissible ambient noise levels for audiometric test rooms.
ANSI/ASA S3.6. (2018). Specification for audiometers.
Bennet-Clark, H. C. (1998). How cicadas make their noise. *Scientific American, 278*(5), 58–61.
Berger, E. H. (2006). Options in defining background noise during audiometric testing. *CAOHC, 18*(1), 1–7.
Berger, E. H., & Killion, M. C. (1989). Comparison of the noise attenuation of three audiometric earphones, with additional data on masking 30 near threshold. *Journal of Acoustical Society of America, 86*(4), 1392–1403.
Clark, J. L., & Roeser, R. J. (1988). Three studies comparing performance of the ER-3A tubephone with the TDH-50P earphone. *Ear and Hearing, 9*(5), 268–274.
Decker, T. N., & Carrell, D. T. (2004). *Instrumentation: An introduction for students in the speech and hearing sciences* (3rd Ed.). Mahwah, NJ: Lawrence Erlbaum Associates.
Egan, J. P., & Hake, H. W. (1950). On the masking pattern of a simple auditory stimulus. *Journal of Acoustical Society of America, 22*, 622–630.
Frank, T. (1994). Ambient noise levels in industrial audiometric test rooms. *American Industrial Hygiene Association Journal; Akron, 55*(5), 433–437. Vol.IssMay 1994).
Frank, T., & Williams, D. L. (1993). Ambient noise levels in audiometric test rooms used for clinical audiometry. *Ear and Hearing, 14*(6), 414–422.
Gross, A. (2005). Insert earphones for occupational hearing conservation testing. *Audiology Today, 17*(6), 21–24.
Hawkins, J. E., & Stevens, S. S. (1950). The masking of pure-tones and of speech by white noise. *Journal of Acoustical Society of America, 22*(1), 6–13.
IEC 60645-1 (2017). Electroacoustics – Audiometric equipment. Part I: Equipment for pure tone and speech audiometry.
IEC 61672-1 (2013). Electroacoustics – Sound level meters. Part 1. Specifications.
ISO 389-1 (2017). Acoustics – Reference zero for the calibration of audiometric equipment. Part 1: Reference equivalent threshold sound pressure levels for pure tones and supra-aural earphones.
ISO 389-2:(1994). (R2020). Acoustics – Reference zero for the calibration of audiometric equipment. Part 2: Reference equivalent threshold sound pressure levels for pure tones and insert earphones.
ISO 389-3 (2016). Acoustics – Reference zero for the calibration of audiometric equipment. Part 3: Reference equivalent threshold vibratory force levels for pure tones and bone vibrators.
ISO 389-4:(1994). (R2021). Acoustics – Reference zero for the calibation of audiometric equipment. Part 4: Reference levels for narrow-band masking noise.
ISO 389-5:(2006). (R2020). Acoustics – Reference zero for the calibration of audiometric equipment. Part 5: Reference equivalent threshold sound pressure levels for pure tones in the frequency range 8 kHz to 16 kHz.
ISO 389-6:(2007). (R2021). Acoustics — Reference zero for the calibration of audiometric equipment. Part 6: Reference threshold of hearing for test signals of short duration.
ISO 389-7:(2019). Acoustics — Reference zero for the calibration of audiometric equipment. Part 7: Reference threshold of hearing under free-field and diffuse-field listening conditions.
ISO 389-8:(2004). (R2017). Acoustics — Reference zero for the calibration of audiometric equipment. Part 8: Reference equivalent threshold sound pressure levels for pure tones and circumaural earphones.
ISO 8253-1: (2010). (R2021). Acoustics — Audiometric test methods — Part 1: Pure-tone air and bone conduction audiometry.
Killion, M., Wilber, L. A., & Gudmundsen, G. I. (1985). Insert earphones for more interaural attenuation. *Hearing instrument, 36*(34), 36.
Lankford, J. E., Perrone, D. C., & Thunder, T. D. (1999). Ambient noise levels in mobile audiometric testing facilities: Compliance with industry standards. *AAOHN Journal, 47*(4), 163–167.
Margolis, R. H., & Madsen, B. (2015). The acoustic test environment for hearing testing. *Journal of American Academy of Audiology, 26*(9), 784–791.

Margolis, R. H., Saly, G. L., & Wilson, R. H. (2022). Ambient noise monitoring during pure-tone audiometry. *Journal of American Academy of Audiology, 33*, 45–56.
Munro, K. J., & Agnew, N. (1999). A comparison of inter-aural attenuation with the Etymotic ER-3A insert earphone and the telephonics TDH-39 supra-aural earphone. *British Journal of Audiology, 33*(4), 259–262.
Munro, K. J., & Contractor, A. (2010). Inter-aural attenuation with insert earphones. *International Journal of Audiology, 49*(10), 799–801.
Oda, D. T. M., Marangoni, A. T., & Gil, D. (2014). Insertion and supra-aural earphones: Clinical assessment in the elderly. *Revista CEFAC, 16*(1), 31–38.
OSHA (2021). Audiometer test rooms. https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.95AppD Accessed on Dec 29, 2021.
Silman, S., Emmer, M. B., Silverman, C. A., & Brody, A. (2022). *Instrumentation for audiology and hearing science: Theory and practice* (2nd Edition). San Diego, CA: Plural Publishing.
Smull, C. C., Madsen, B., & Margolis, R. H. (2019). Evaluation of two circumaural earphones for audiometry. *Ear and Hearing, 40*(1), 177–183.
Stangl, S. (2015). *Accuracy of audiometry conducted outside of a sound-treated booth*. Presentation at Vanderbilt University. https://coalitionforglobalhearinghealth.org/wp-content/uploads/2015/12/Stangl-Accuracy-of-Audiometry-Conducted-Outside-of-a-Sound-Treated-Booth.pdf.
Swanepoel, D. W., Matthysen, C., Eikelboom, R. H., Clark, J., & Hall, J. W. (2015). Pure-tone audiometry outside a sound booth using earphone attenuation, integrated noise monitoring, and automation. *International Journal of Audiology, 54*, 777–785.
Werner, L. A., & Bargones, J. Y. (1991). Sources of auditory masking in infants: Distraction effects. *Perception and Psychophysics, 50*(5), 405–412.
Wilber, L. A., & Burkard, R. (2015). Calibration. *Katz eds: Handbook of clinical audiology* (7th Ed, pp. p9–27). Philadelphia, PA: Wolters Kluwer Health.
Yacullo, W. A. (2015). *Clinical masking* (7th Ed., pp. p77–112). Philadelphia, PA: Wolters Kluwer HealthHandbook of Clinical Audiology. In Katz Eds.